STA4001 2020 Fall midterm WANG Youmo 12009018

1. Solution. Model the states to three conditions, and form a DTMC.

Excellent: $X_n=1$; Workable: $X_n=2$; Failed: $X_n=3$.

Do nothing: $U_n=1$; Operational mistakes: $U_n=2$; Operational failure: $U_n=3$; Repair: $U_n=4$.

DTMC: $X_{n+1}=f(X_n, U_{n+1})$, $n\in\mathbb{N}^+$

X_n	U_{n+1}	X_{n+1}
1	1	1
1	2	2
1	3	3
2	1	2
2	3	3
3	4	1

(a). Modelled the system DTMC as above.

The one-step transition matrix is: $P=\begin{bmatrix}0.85 & 0.1 & 0.05\\ 0 & 0.7 & 0.3\\ 1 & 0 & 0\end{bmatrix}$

(b) Stationary distribution $\pi=[\pi_1,\pi_2,\pi_3]$. Solve $\begin{cases}\pi=\pi P\\ \sum_{i\in S}\pi_i=1\end{cases}$.

We derive $\pi=\left[\frac{60}{89},\frac{20}{89},\frac{9}{89}\right]$. The long-run fraction of days spent on repairing: $1\times\pi_3=\frac{9}{89}$.

(c) Value function $g(X_n)$, $X_n=1,2,3$. $g=[1000,300,-500]^T$.

The long-run average daily profit is $\pi g=\frac{61500}{89}$.

2. Solution. $C_v=10$ $C_p=30$ $C_s=5$ $q=8$ $D\sim$ Uniform$(5,10)$

(a) profit $=C_p\min(q,D)-C_vq+C_s(q-D)^+=C_p[q-(q-D)^+]-C_vq+C_s(q-D)^+$

$=(C_s-C_p)(q-D)^+ +(C_p-C_v)q=-25(8-D)^+ +160$.

$E(\text{profit})=-25E[(8-D)^+]+160$, where $E[(8-D)^+]=\int_5^8(8-x)\cdot\frac{1}{5}dx=0.9$.

$\therefore E(\text{profit})=-25\times0.9+160=137.5$; $\text{Var}(\text{profit})=E(\text{profit}^2)-[E(\text{profit})]^2$, where

$E(\text{profit}^2)=E\{625[(8-D)^+]^2\}-E[8000(8-D)^+]+25600$, and $E\{[(8-D)^+]^2\}=\int_5^8(8-x)^2\cdot\frac{1}{5}dx=\frac{9}{5}=1.8$

$=625\times1.8-8000\times0.9+25600=19525$, $\text{Var}(\text{profit})=19525-18906.25=618.75$.

(b) $\mu=137.5$ $\sigma^2=618.75$, $n=100$; $13750-\sqrt{100}\times1.96\times\sqrt{618.75}\le\sum_{i=1}^{100}X_i\le13750+\sqrt{100}\times1.96\times\sqrt{618.75}$

Hence, the 95% confidence interval is $[13750-19.6\sqrt{618.75},\ 13750+19.6\sqrt{618.75}]$.

(c) $F(x)=P(D\le x)=\int_5^x\frac{1}{5}dx=\frac{1}{5}(x-5)$ $[5\le x\le10]$. By corollary, x^* is the smallest x satisfies $F(x^*)\ge\frac{C_p-C_v}{C_p-C_s}=\frac{20}{25}=0.8$. We conclude $x^*=9$, i.e., optimal q.

3. Solution $T_i=\min\{n\ge1\mid X_n=i\}$ be the first time to reach state i.

(a) Suppose $q_i=P(T_2<T_4\mid X_0=i)$. We list the condition:

$$\begin{cases}q_1=\frac{1}{2}\times1+\frac{1}{2}q_5\\ q_5=\frac{1}{2}\times0+\frac{1}{2}q_1\\ q_4=\frac{1}{2}q_3+\frac{1}{2}q_5\\ q_3=\frac{1}{2}\times1+\frac{1}{2}\times0\\ q_2=\frac{1}{2}q_3+\frac{1}{2}q_1\end{cases}\Longrightarrow\begin{cases}q_1=\frac{3}{3}\\ q_2=\frac{7}{12}\\ q_3=\frac{1}{2}\\ q_4=\frac{5}{12}\\ q_5=\frac{1}{3}\end{cases}$$

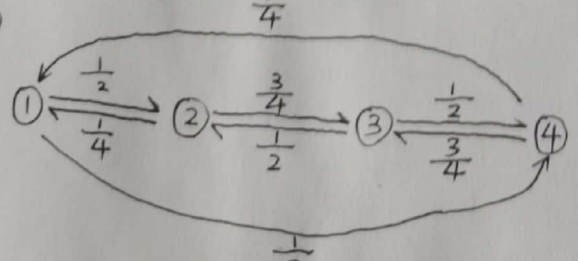
(b) Similarly, suppose N_i be the expected number of steps to reach 4 from state i.

$$\begin{cases}N_1=1+\frac{1}{2}N_5+\frac{1}{2}N_2\\ N_2=1+\frac{1}{2}N_1+\frac{1}{4}N_3\\ N_3=1+\frac{1}{2}\times0+\frac{1}{2}N_2\\ N_5=1+\frac{1}{2}\times0+\frac{1}{2}N_1\end{cases}\text{, and } N_4=0\Longrightarrow\begin{cases}N_1=6\\ N_2=6\\ N_3=4\\ N_4=0\\ N_5=4\end{cases}$$

that is, $P(T_2<T_4\mid X_0=j)=\left[\frac{3}{3},\frac{7}{12},\frac{1}{2},\frac{5}{12},\frac{1}{3}\right]^T$, and $E_1(T_4)=N_1=6$.

4. Solution. $X=\{X_n: n=0,1,2,\cdots\}$ DTMC, $S=\{1,2,3,4\}$ $P=\begin{bmatrix}0 & \frac{1}{2} & 0 & \frac{1}{2}\\ \frac{1}{4} & 0 & \frac{3}{4} & 0\\ 0 & \frac{1}{2} & 0 & \frac{1}{2}\\ \frac{1}{4} & 0 & \frac{3}{4} & 0\end{bmatrix}$

(a)



(b) Compute

$$P^2=\begin{bmatrix}\frac{1}{4} & 0 & \frac{3}{4} & 0\\ 0 & \frac{1}{2} & 0 & \frac{1}{2}\\ \frac{1}{4} & 0 & \frac{3}{4} & 0\\ 0 & \frac{1}{2} & 0 & \frac{1}{2}\end{bmatrix}$$

Therefore, $P(X_2=4\mid X_0=2)=\{P^2\}_{2,4}=\frac{1}{2}$.

(c) $P\{X_2=2, X_4=4, X_5=1 \mid X_0=2\} = P(X_5=1 \mid X_0=2, X_2=2, X_4=4)\, P\{X_4=4 \mid X_0=2, X_2=2\}\, P\{X_2=2 \mid X_0=2\}$

(Markov Property) $= P(X_5=1 \mid X_4=4)\, P(X_4=4 \mid X_2=2)\, P(X_2=2 \mid X_0=2)$.

$= P_{4,1} \cdot \{P^2\}_{2,4} \cdot \{P^2\}_{2,2} = \frac{1}{4} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$.

(d) We may find that $P^{2n} = P^2$, $P^{2n-1} = P$ $(n \in \mathbb{N}^+)$. Hence, $d(1)=2$, implied by $\{P^{2n}\}_{1,1} = \frac{1}{4} > 0$, and $\{P^{2n-1}\}_{1,1} = 0$. Namely, $d(i)=2$, as $\{P^{2n}\}_{i,i} > 0$, $\{P^{2n-1}\}_{i,i} = 0$.

(e) Since this finite DTMC is irreducible, it is positive recurrent. Hence it has a unique stationary distribution. Check the correctness of $(\frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{1}{4})$. Conclusion.

(f) We've already computed that $P^{2n} \neq P^{2n-1}$, i.e., $\{P^{2n}\}_{1,4} = 0 \neq \frac{1}{2} = \{P^{2n-1}\}_{1,4}$, $\forall n \in \mathbb{N}^+$.

Therefore, $\lim_{n \to \infty} \{P^n\}_{1,4} = \frac{1}{4}$ cannot hold.

(g) $\pi_1 = \frac{1}{E_1(T_1)} \Rightarrow E_1(T_1) = 8$.

5. Solution. We can modify the two-dimensional random walk so that it satisfies the requirements. Add T(transient), P(positive recurrent) into the state space S to construct a state space S', that is: $S' = S \cup S_p \cup T = \mathbb{Z} \cup \{P\} \cup \{T\}$.

① All of states $i \in S$, state i can go to state $i-1$, state i itself, or state $i+1$, each one with probability $\frac{1}{3}$.

② For state $t \in T$, it can go to state 0, state T, or state P, each with probability $\frac{1}{3}$.

③ For state $p \in P$, it can only go to itself with probability 1.